

Digital Electronics and Logic Design (ELE1204)

BY

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Chapter 1

- **Introduction to Digital Systems**
 - Analog vs. Digital Signals
 - Applications in computer systems
- **Number Systems and Codes**
 - Binary number system and representation
 - Octal, Decimal, Hexadecimal and Binary Conversions
 - BCD (Binary Coded Decimal)
 - Binary Arithmetic (Addition, Subtraction, Multiplication, Division)
- **Boolean Algebra**
 - Laws and Theorems
 - Duality principle
 - Truth tables

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- **Logic Gates**
 - Basic gates (AND, OR, NOT)
 - Universal gates (NAND, NOR)
 - XOR and XNOR gates

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Analog vs. Digital Signals

analog signal



digital signal



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Analog vs. Digital Signals

Which of the following involve analog quantities and which involve digital quantities?

- (a) Elevation using a ladder
- (b) Elevation using a ramp
- (c) Current flowing from an electrical outlet through a motor
- (d) Height of a child measured by a yard stick ruler
- (e) Height of a child measured by putting a mark on the wall
- (f) Amount of rocks in a bucket
- (g) Amount of sand in a bucket
- (h) Volume of water in a bucket

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- Analog vs. Digital Signals
 - **Analogue signal** is a type of signal that represents continuous data using a continuous range of values
 - It can take on any value within a certain range
 - Represented by a smooth and continuous graph

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- Analog vs. Digital Signals
 - **Digital signal** is a type that represents data as a sequence of discrete values (usually in binary digits).
 - Signals take on only a finite set of values
 - Represented by step/pulse graphs
 - Mostly used in telecommunications.

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● Applications of Digital systems

- Electric stoves, traffic lights, Digital signature pads, motion sensors,
- Fire alarms, Electronic safes, Automobile systems, washing machines,
- parking IOT ticket machines, printers, photocopy, fax machines and scanners,
- Wi-Fi routers, vending machines, Digital cameras.
- ...etc...

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- Analog vs. Digital Signals

Basis	Analog Signal	Digital Signal
Definition	Analog signals represent continuous variations in magnitude over time.	Digital signals are Discrete and quantized, with specific values.
Signal Type	Continuous waveforms	Discrete Signals
Processing	Requires complex processing for manipulation.	Easier to process and manipulate digitally.
Storage	Less efficient for storage due to continuous nature.	More efficient for storage due to discrete values.
Bandwidth	Typically requires more bandwidth.	Requires less bandwidth for transmission.
Examples	Analog audio signals, analog radio waves, Human voice, etc.	Digital audio signals, digital data streams, computers, etc.
Errors	Susceptible to noise and distortion	More resistant to noise and distortion
Circuit Component	Amplifiers, filters, continuous-wave oscillators	Microprocessors, binary counters, logic gates
Signal Values	Infinite range of values	Limited to discrete values
Conversion	No conversion required	Analog-to-digital conversion (ADC) required
Applications	Analog signals are used in electric fan, landlines, radio frequency communications, etc.	Digital signals are used in computers, smartphones, digital sensors, digital imaging, etc.

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- Advantages of Digital Signals

1. Digital systems are generally easier to design. The circuits used in digital systems are switching circuits, where exact values of voltage or current are not important, only the range (HIGH or LOW) in which they fall.
2. Information storage is easy. This is accomplished by special devices and circuits that can latch onto digital information and hold it for as long as necessary, and mass storage techniques that can store billions of bits of information in a relatively small physical space. Analog storage capabilities are, by contrast, extremely limited.
3. Accuracy and precision are easier to maintain throughout the system. Once a signal is digitized, the degree to which it deteriorates is predictable and more easily contained within acceptable limits. In analog systems, the voltage and current signals tend to be distorted by the effects of temperature, humidity, and component tolerance variations in the circuits that process the signal.

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- Advantages of Digital Signals

4. Operations can be programmed. It is fairly easy to design digital systems whose operation is controlled by a set of stored instructions called a program. Analog systems can also be programmed, but the variety and the complexity of the available operations are severely limited.

5. Digital circuits are less affected by noise. Spurious fluctuations in voltage (noise) are not as critical in digital systems because the exact value of a voltage is not important, as long as the noise is not large enough to prevent us from distinguishing a HIGH from a LOW.

6. More digital circuitry can be fabricated on IC chips. It is true that analog circuitry has also benefited from the tremendous development of IC technology, but its relative complexity and its use of devices that cannot be economically integrated (high-value capacitors, precision resistors, inductors, transformers) have prevented analog systems from achieving the same high degree of integration.

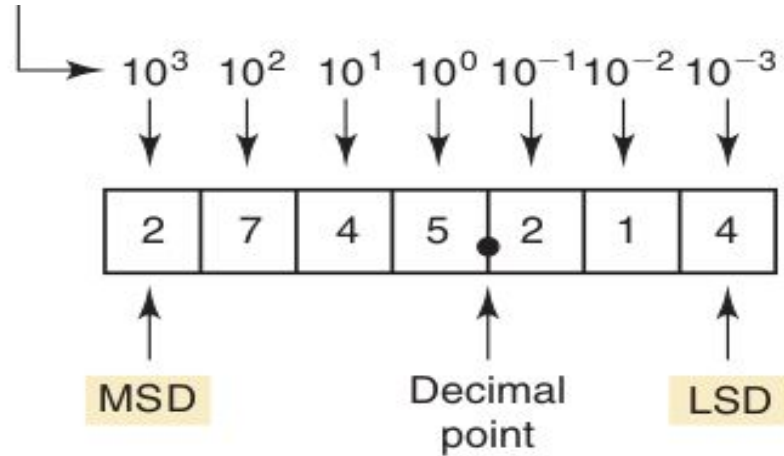
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- Limitations of Digital Signals
 1. The real world is analog and digitizing always introduces some error.
 2. Processing digitised signals takes time

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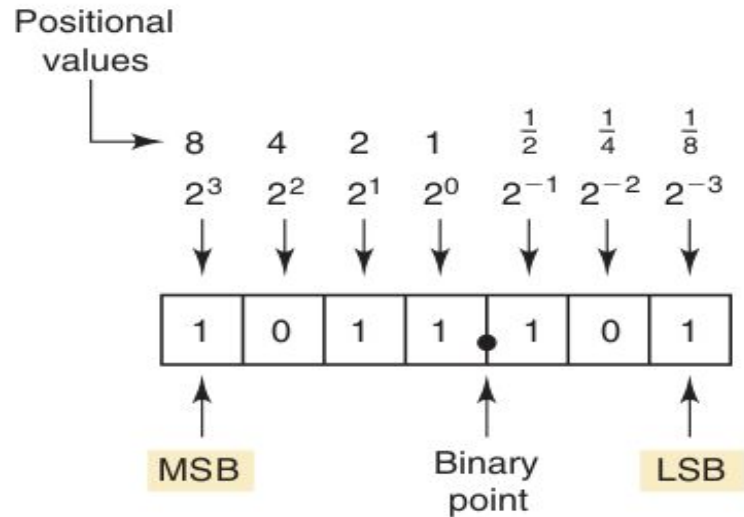
- **Digital Number systems.**

- The most common are the decimal, binary, and hexadecimal systems
- **Decimal systems**
 - $27.35 = (2 \times 10) + (7 \times 1) + (3 \times 0.1) + (5 \times 0.01)$



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- Digital Number systems.
 - Binary systems



Example: $1011.101_2 = (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$
 $+ (1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3})$
 $= 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125$
 $= 11.625_{10}$

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- Octal and Hexadecimal

Binary to Octal Table

Binary (Base 2)	Octal (Base 8)	Binary (Base 2)	Octal (Base 8)
000	0	100	4
001	1	101	5
010	2	110	6
011	3	111	7

Binary to Hexadecimal Table

Binary (Base 2)	Hexadecimal (Base 16)	Binary (Base 2)	Hexadecimal (Base 16)
0000	0	1000	8
0001	1	1001	9
0010	2	1010	A
0011	3	1011	B
0100	4	1100	C
0101	5	1101	D
0110	6	1110	E
0111	7	1111	F

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- **BCD (Binary Coded Decimal)**

- Is another process for converting decimal numbers into their binary equivalents.
- It is a form of binary encoding where each digit in a decimal is represented in the form of bits
- Encoding is done wither 4-bit or 8-bit (4-bit is preferred).

<i>BCD Code</i>	<i>Decimal Number</i>	<i>BCD Code</i>	<i>Decimal Number</i>
0000	0	0101	5
0001	1	0110	6
0010	2	0111	7
0011	3	1000	8
0100	4	1001	9

1. Convert (123)₁₀ in BCD

2. Convert (324)₁₀ in BCD

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- Binary Arithmetic

Input A	Input B	Sum (S) A+B	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Input A	Input B	Subtract (S) A-B	Borrow (B)
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

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- Binary Arithmetic

Input A	Input B	Multiply (M) $A \times B$
0	0	0
0	1	0
1	0	0
1	1	1

Input A	Input B	Divide (D) A/B
0	0	Not defined
0	1	0
1	0	Not defined
1	1	1

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- **Boolean Algebra and Logic Gates**
 - **Boolean algebra** is a means for expressing the relationship between a logic circuit's inputs and outputs. The inputs are considered logic variables whose logic levels at any time determine the output levels.
 - In Boolean algebra there are only three basic operations: OR, AND, and NOT (Logic operations)
 - Digital circuits called logic gates can be constructed from diodes, transistors, and resistors connected so that the circuit output is the result of a basic logic operation

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- Boolean Algebraic Laws and Theorems

Name	AND form	OR form
Identity law	$1A = A$	$0 + A = A$
Null law	$0A = 0$	$1 + A = 1$
Idempotent law	$AA = A$	$A + A = A$
Inverse law	$A\bar{A} = 0$	$A + \bar{A} = 1$
Commutative law	$AB = BA$	$A + B = B + A$
Associative law	$(AB)C = A(BC)$	$(A + B) + C = A + (B + C)$
Distributive law	$A + BC = (A + B)(A + C)$	$A(B + C) = AB + AC$
Absorption law	$A(A + B) = A$	$A + AB = A$
De Morgan's law	$\overline{AB} = \bar{A} + \bar{B}$	$\overline{A + B} = \bar{A}\bar{B}$

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- **Duality Principle**

- Primarily utilized in the process of proving a variety of theorems that can be found in Boolean algebra
- States that in a two-valued boolean algebra, the dual of an algebraic expression can be obtained by exchanging all of the OR and AND operators and then replacing 1 with 0 and 0 with 1

- **Steps used in the Duality Theorem**

- Change each AND operation to an OR operation.
- Change each OR operation to an AND operation.
- Replace 0 with 1.
- Replace 1 with 0.

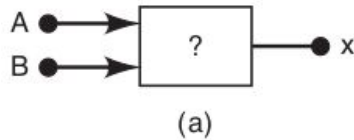
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- **Truth Tables**

- **A truth table** is a means for describing how a logic circuit's output depends on the logic levels present at the circuit's inputs

Diagram illustrating a logic circuit with two inputs (A, B) and one output (x). The inputs are labeled "Inputs" and the output is labeled "Output".

A	B	x
0	0	1
0	1	0
1	0	1
1	1	0



A	B	C	x
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

(b)

A	B	C	D	x
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

(c)

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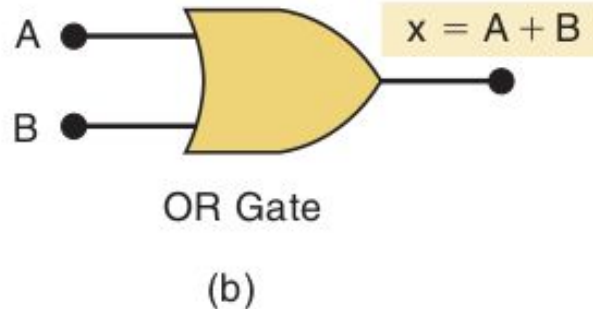
- **Operation with basic gates**

- **OR gates:** Output is HIGH (logic 1) if either input A or B or both are at a logic 1 level.
- The OR operation produces a result (output) of 1 whenever any input is a 1. Otherwise the output is 0.
- An OR gate is a logic circuit that performs an OR operation on the circuit's inputs.
- The expression $x = A + B$ is read as “x equals A OR B.”

OR

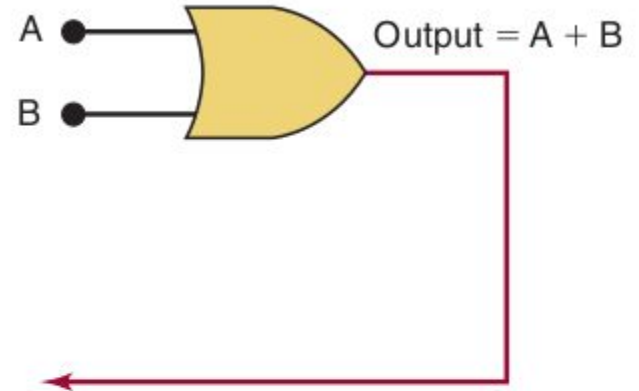
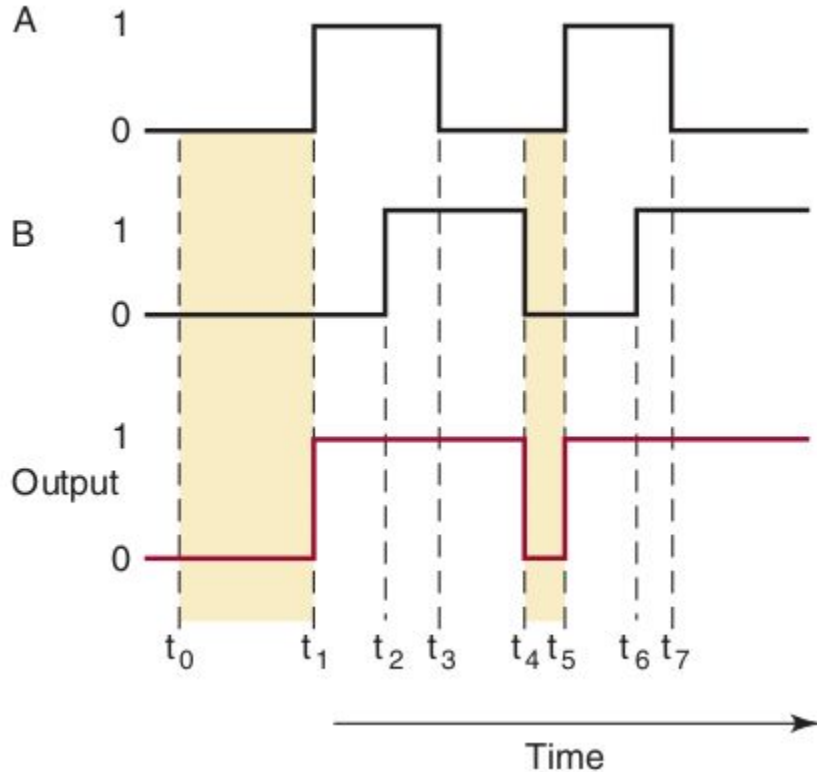
A	B	$x = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

(a)



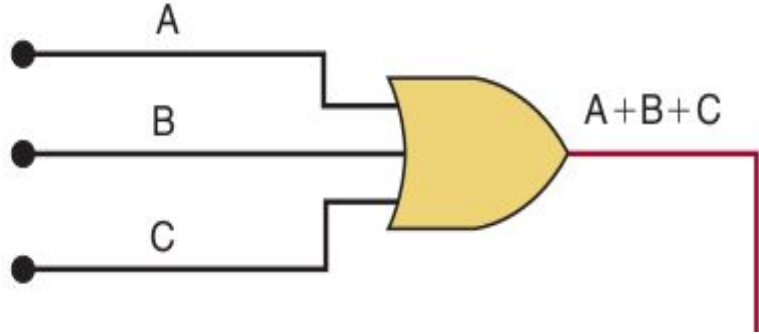
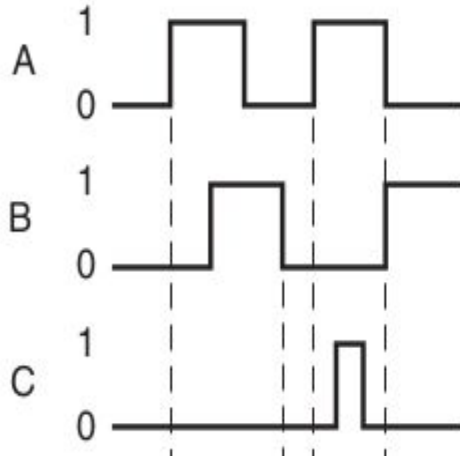
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- Operation with OR gates



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- Operation with OR gates (3 Inputs)



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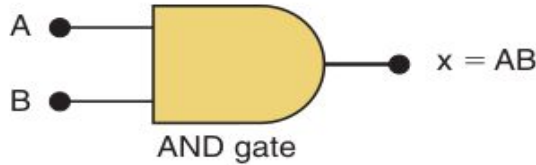
- **Operation with basic gates**

- **AND gates:** Output is HIGH (logic 1) if both input A and B or both are at a logic 1 level.
- The AND operation produces a result (output) of 1 whenever both inputs are 1. Otherwise the output is 0.
- An AND gate is a logic circuit that performs an AND operation on the circuit's inputs.
- The expression $x = A \cdot B$ is read as “x equals A AND B,”

AND

A	B	$x = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

(a)



(b)

The Boolean expression for the AND operation is

$$x = A \cdot B$$

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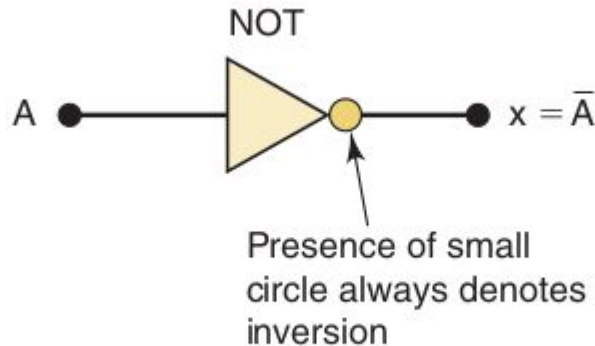
- Operation with basic gates
 - NOT gates

$0 = \bar{1}$ because 0 is not 1

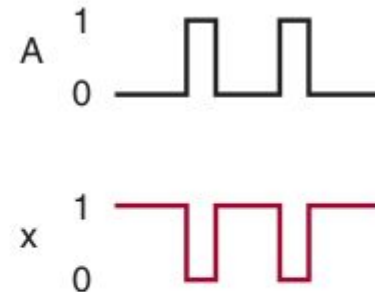
$1 = \bar{0}$ because 1 is not 0

NOT		
A		$x = \bar{A}$
0		1
1		0

(a)



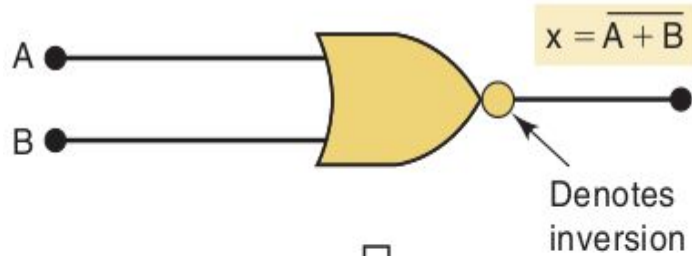
(b)



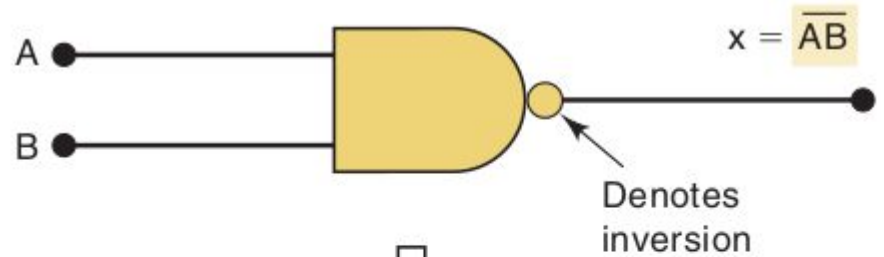
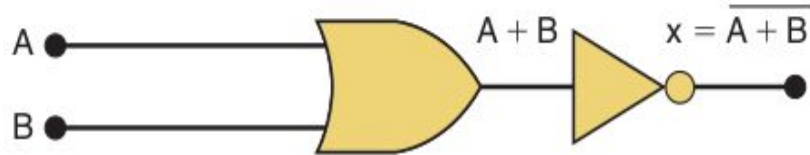
(c)

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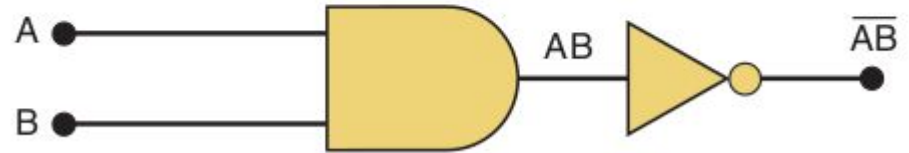
- Operation with Universal gates
 - NOR and NAND gates



(a) ↓



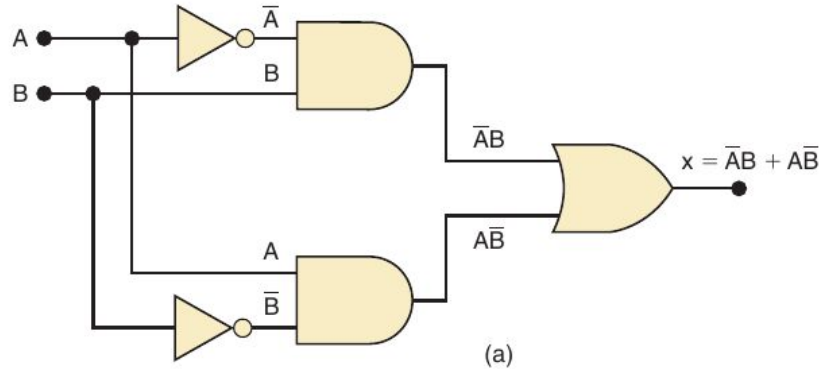
(a) ↓



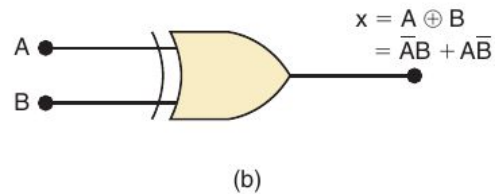
(b)

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- Operation with other gates
 - XOR gates

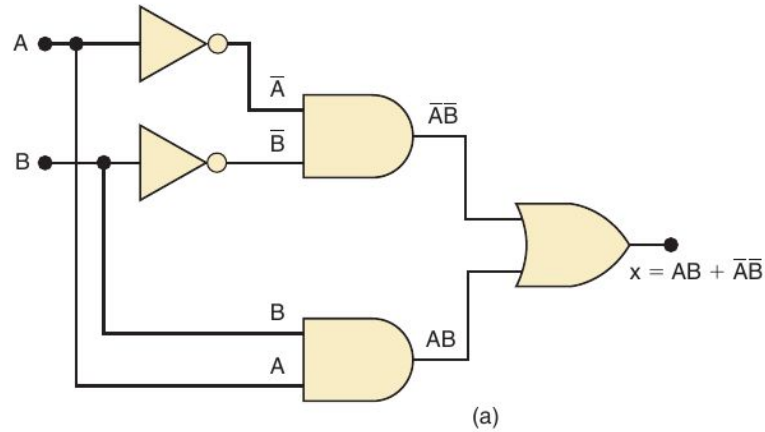


A	B	x
0	0	0
0	1	1
1	0	1
1	1	0

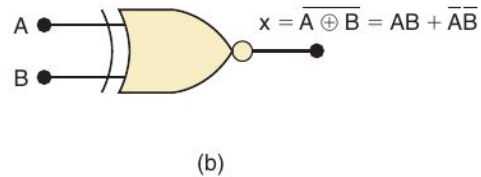


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- Operation with other gates
 - XNOR gates

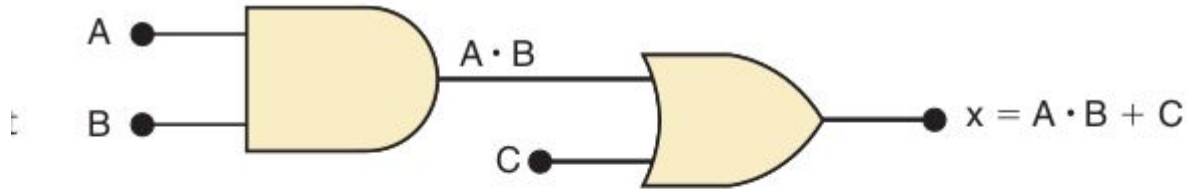


A	B	x
0	0	1
0	1	0
1	0	0
1	1	1

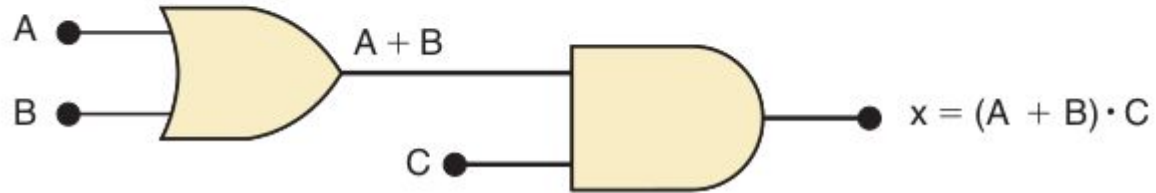


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- **Describing Logic circuits algebraically**
 - Translate logic diagrams into Boolean algebraic expressions

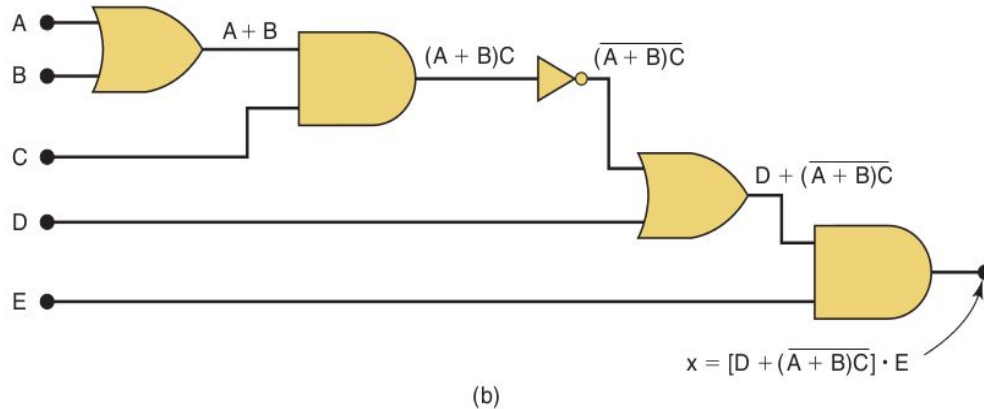
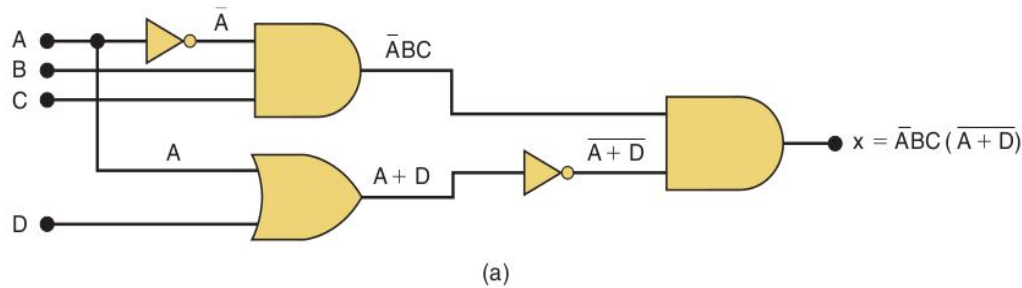


(a)



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- Describing Logic circuits algebraically
 - Translate logic diagrams into Boolean algebraic expressions



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- **Reading and Practice assignment**

- Further reading: Read Chapters 2 and 3 - Digital Electronics 12th Edition by Ronald Tocci
- Further reading: Read Chapter 1 - Digital Design 5th Edition by Morris Mano
- Exercises: Attempt End-of-Chapter-1 Problems: 1.1 - 1.12 - Digital Design 5th Edition by Morris Mano

Thank you